

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in an orderly fashion maintaining all mathematical norms and rules.
- Start working right away based on whatever you know. **Do not wait for the last moment and ask for a time extension.**

1. In the classes, there are three forms of floating number representation,

$$\begin{aligned}\text{Standard Form} &: F = \pm(0.d_1d_2d_3\cdots d_m)_\beta \beta^e, \\ \text{Denormalized Form} &: F = \pm(1.d_1d_2d_3\cdots d_m)_\beta \beta^e, \\ \text{Normalized Form} &: F = \pm(0.1d_1d_2d_3\cdots d_m)_\beta \beta^e,\end{aligned}$$

where $d_i, \beta, e \in \mathbb{Z}$, $0 \leq d_i \leq \beta - 1$ and $e_{\min} \leq e \leq e_{\max}$. Now, let's take, $\beta = 2$, $m = 4$ and $e \in \{-2, -1, 0, 1, 2\}$. Based on these, answer the following:

- (3 marks) How many numbers in total can be represented by this system? Find this separately for each of the three forms above. Ignore negative numbers.
- (3 marks) For each of the three forms, find the smallest, positive number and the largest number representable by the system.
- (2 marks) For the IEEE standard (1985) for double-precision (64-bit) arithmetic, find the smallest, positive number and the largest number representable by a system that follows this standard. Do not find their decimal values, but simply represent the numbers in the following format:

$$\pm(0.1d_1\dots d_m)_\beta \cdot \beta^{e-\text{exponentBias}}$$

Be mindful of the conditions for representing $\pm\text{inf}$ and ± 0 in this IEEE standard.

- (2 marks) In the above IEEE standard, if the exponent bias were to be altered to $\text{exponentBias} = 500$, what would the smallest, positive number and the largest number be? Write your answers in the same format as in part (c). Note that the conditions for representing $\pm\infty$ and ± 0 are still maintained as before.

2. Given a system parameterized by $\beta = 2$, $m = 3$, and $e_{\min} = -1 \leq e \leq e_{\max} = 2$ where $e \in \mathbb{Z}$. For this system,

- (4 marks) find the floating-point representation of the numbers $(6.25)_{10}$ and $(6.875)_{10}$ in the Normalized Form. That is, find $\text{fl}[6.25]$ and $\text{fl}[6.875]$.
- (4 marks) what are the rounding errors δ_1, δ_2 in part (a)?
- (2 marks) can the values $(6.25)_{10}$ and $(6.875)_{10}$ be represented in the Denormalized Form? If so, find the floating-point representations. If not, then concisely explain why?

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules.
- Start working right away based on whatever you know. **Do not wait for the last moment and ask for time extension.**

1. The following nodes come from the function $f(x) = \ln(5x + 9)$:

x	$f(x)$
-0.5	1.87
0	2.20
0.5	2.44

- (4 marks) Using Newton's divided difference method, find the equation of a second degree polynomial which fits the above data points.
 - (5 marks) Expand the function $f(x) = \ln(5x + 9)$ using Taylor Series, centered at 0. Include till the x^2 term of the Taylor series.
 - (1 mark) Should the equation which you found in part (a) and part (b) match? Comment on why, or why not.
2. Consider the following nodes:

x	$f(x)$
0	5
3	9.5
6	5

- (1 mark) If an equation of a polynomial which fits through the above nodes is found using both the Vandermonde Matrix approach and the Lagrange approach, will both the equations match?
 - (7 marks) Find the equation of a polynomial which fits through the above nodes using the Vandermonde matrix approach.
 - (7 marks) Find the equation of a polynomial which fits through the above nodes using the Lagrange approach.
- 3.
- (5 marks) Compute the upper bound of error using Cauchy's Theorem for $f(x) = \sin^2(x/2)$ with nodes $\{-\pi/3, 0, \pi/3\}$ within the interval $[-1.2, 1.2]$.

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules.
- Start working right away based on whatever you know. **Do not wait for the last moment and ask for time extension.**

1. A function is given by $f(x) = 6e^{-5x}$. Now answer the following:

- (3 marks) Approximate the derivative of $f(x)$ at $x_0 = 0.2$ with step size $h = 0.5$ using the central difference method up to 6 significant figures.
- (3 marks) Approximate the derivative of $f(x)$ at $x_0 = 0.2$ with step size $h = 0.5$ using the forward difference method up to 6 significant figures.
- (4 marks) Calculate the truncation error of $f(x)$ at $x_0 = 2$ using $h = 1, 0.1, 0.01, 0.0001$ in the above mentioned two methods.
- (5 marks) Compute $D_{0.5}^{(1)}$ at $x_0 = 0.2$ using Richardson extrapolation method up to 6 significant figures and calculate the truncation error.

2. During the class, we derived in detail the first order Richardson extrapolated derivative, by using $h \rightarrow h/2$,

$$D_h^{(1)} \equiv f'(x_0) - \frac{h^4}{480} + O(h^6).$$

- (8 marks) Using $h \rightarrow h/2$, derive the expression for $D_h^{(2)}$ which is the second order Richardson extrapolation.
- (7 marks) Now starting from the definition of D_h and using $h \rightarrow h/3$, derive the expression for $D_h^{(1)}$.

Instructions for preparing the
solution script:

- Write your name, ID#, and Section number clearly in the very front page.
 - Write all answers sequentially.
 - Start answering a question (not the part of the question) from the top of a new page.
 - Write legibly and in orderly fashion maintaining all mathematical norms and rules.
 - Start working right away based on whatever you know. **Do not wait for the last moment and ask for time extension.**
-

1. Consider the function, $f(x) = x^3 - x^2 - 9x + 9$. Answer the following:
 - (a) (3 marks) **State** the exact roots of $f(x)$.
 - (b) (3 marks) **Construct** three different fixed point functions $g(x)$ such that $f(x) = 0$. (Make sure that one of the $g(x)$'s that you constructed converges to at least a root).
 - (c) (3 marks) **Find** the convergence rate/ratio for each $g(x)$ constructed in the previous part and also **find** which root it is converging to.
 - (d) (6 marks) **Find** the approximate root, x_* , of the above function using fixed point iterations up to 4 significant figures within the error bound of 1×10^{-3} using $x_0 = 0$ and any fixed point function $g(x)$ from part(b) that converges to the root(s).
2. (15 marks) Use Newton's method to **find** the root, x_* , of the equation, $f(x) = x^2 e^{-x} - 0.6$, up to machine epsilon of 1×10^{-4} starting with $x_0 = 0.2$.

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
 - Write all answers sequentially.
 - Start answering a question (not the part of the question) from the top of a new page.
 - Write legibly and in orderly fashion, maintaining all mathematical norms and rules.
 - Start working right away based on whatever you know. **Do not wait for the last moment and ask for time extension.**
-

1. Consider the following linear system,

$$\begin{aligned}2x_1 + x_2 - x_3 + 2x_4 &= 5 \\4x_1 + 5x_2 - 3x_3 + 6x_4 &= 9 \\-2x_1 + 5x_2 - 2x_3 + 6x_4 &= 4 \\4x_1 + 11x_2 - 4x_3 + 8x_4 &= 2.\end{aligned}$$

- (3 marks) Explain how to find this system has a unique solution or not.
- (2 mark) Write down the Augmented matrix for the system (*i.e.* write down $\text{Aug}(A)$).
- (4 marks) Find the upper triangular matrix U .
- (6 marks) Using the Gaussian elimination method, find the solution of the linear system. Show your works to get full credit.

2. A linear system is described by the following equations

$$\begin{aligned}x_1 + x_2 + x_3 &= 6 \\2x_1 + 3x_2 + 4x_3 &= 20 \\3x_1 + 4x_2 + 2x_3 &= 17.\end{aligned}$$

- (3 marks) Construct the Frobenius matrices $F^{(1)}$ and $F^{(2)}$ from this system.
- (2 marks) Compute the unit lower triangular matrix L .
- (5 marks) Now find the solution of the linear system using LU decomposition method. Use the unit lower triangular matrix found in the previous question.

3. A linear system is defined by three independent variables that obey the following equations

$$\begin{aligned}3x_1 + 3x_2 + 4x_3 &= 1 \\-x_3 &= 6 \\-x_2 + 3x_3 &= 4.\end{aligned}$$

- (2 marks) Explain why the Gaussian elimination method fails to solve the system.
- (3 marks) State how we can remove the problem and solve the system by Gaussian elimination method.